INCH-POUND

MIL-P-14824B (AR)
08 MARCH 1989
SUPERSEDING
MIL-P-14824A (MU)
13 August 1976

MILITARY SPECIFICATION

PROJECTILE, H.E., 155MM, M107 METAL PARTS ASSEMBLY

This specification is approved for use within the US Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 SCOPE. This specification is the applicable detail specification for projectile, H.E., 155MM, M107 Metal Parts Assembly.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

TT-C-490	Cleaning Method and Pre-treatment of Ferrous Surfaces for Organic Coatings

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions, and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1320

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:51Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-A-2550 - Ammunition, General Specification for
MIL-S-10520 - Steel, Forging for Projectile Stock
MIL-S-11310 - Steel Bars, Carbon, Hot Rolled, for Cold Shaping, Including Cold Extrusion
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

FEDERAL

FED-STD-151 - Metal Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-410 - Nondestructive Testing Personnel Qualification of Inspection Personnel
MIL-STD-1168 - Lot Numbering of Ammunition

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

PRODUCT AND PACKAGING DRAWINGS

9362569 - Packing & Marking for Pallet, Wood Ammo for PJCTL 155MM
10535925 - Projectile, H.E., 155MM, M107 Metal Parts Assembly

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:51Z
Check the source to verify that this is the current version before use.

INSPECTION EQUIPMENT DRAWINGS

8593417	- Band Seating Testing Machine
8593429	- Band Seating Testing Machine, Indentor and Test Data
9280419	- Envelope Drawing 155MM, M107 Projectile Ultrasonic Equipment
9280420	- Ultrasonic Test Calibration Standard 155MM, M107 Projectile

PUBLICATIONS

Frankford Arsenal
Operating Instruction for use of Detroit Testing Machine for Band Seat Inspection

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E8	- Tension Testing of Metallic Materials
ASTM E10	- Brinell Hardness of Metallic Materials
ASTM E103	- Rapid Indentation Hardness Testing of Metallic Materials
ASTM B-117	- Salt Spray (Fog) Testing
ASTM A-711	- Carbon Steel Blooms, Billets and Slabs for Forging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets of MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:51Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 General. Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications and the following:

3.2 Steel quality. The item contractor shall establish procedures to provide a forge quality steel when procuring steel in accordance with ASTM A711, i.e., the steel shall be free of pipe, porosity and heavy segregation. The manufacturer will provide a certification to cite the specific revisions of the industry standard under which the steel was manufactured. Material procured in accordance with MIL-S-10520 or MIL-S-11310 shall comply with the requirements of the applicable specifications.

3.3 Heat treatment or stress relief. Heat treatment or stress relief applied to any component part shall be uniform throughout and shall be so conducted as to produce in each component the mechanical properties specified on the applicable drawing.

3.3.1 Hot forge heat treat method. Prior to finish machining, projectile bodies manufactured from forgings shall be subjected to heat treatment to attain the required mechanical properties. The heat treatment shall consist of heating above the critical temperature, not to exceed 1600 degrees Fahrenheit (^{O}F), followed by an oil quench. After quenching, the body forgings shall be tempered without delay at a temperature above 750 degrees Fahrenheit, suitable to produce required mechanical properties. After removal from the tempering furnace, the body forgings shall be slowly cooled in air (forced air permitted) to a temperature of not more than 700 degrees Fahrenheit, after which cooling may be accelerated by water spray.

3.3.2 Cold shaping and hot cup - cold draw methods. Cold working shall be so accomplished as to produce in the projectile body the mechanical properties specified. The projectile body shall be stress relieved at a temperature of 700 degrees Fahrenheit, minimum, for a minimum of 30 minutes following the final forming operation during which time plastic flow takes place.

3.4 Mechanical properties.

3.4.1 Body. Mechanical properties specified on the applicable drawing shall apply throughout the component.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:51Z
Check the source to verify that this is the current version before use.

3.4.2 Rotating band. After application and subsequent removal the rotating band shall have the following mechanical properties:

Tensile Strength - 30,000 lbs per square inch (psi) minimum (min)
Elongation - 25 percent in one inch of gage length, minimum

3.5 Banding.

3.5.1 Band seat. Prior to banding, the band seat shall be clean and free from oil, grease, dirt, rust or other foreign matter. Shot or sandblasting shall not be permitted for cleaning the band seat after knurling.

3.5.2 Band blank - (Gilding Metal). The rotating band blank shall be in the form of a solid ring, annealed if necessary to comply with the applicable specification, except that where hot banding at not less than 800 degrees Fahrenheit (^{O}F) no prior annealing is necessary.

3.5.3 Application. The rotating band shall be applied after final heat treatment of the body, by carefully pressing the band blank into the band seat so as to make contact with the sidewalls for a minimum of eight-tenths of the band seat depth. The band shall be applied in such manner as not to distort the body wall. There shall be no gap between the rotating band and the sidewalls of the band seat, after the rotating band has been machined to final dimension.

3.5.3.1 Application hot or cold. The bands may be applied by either hot or cold but if applied hot, the bands shall not be heated above 1600 degrees Fahrenheit to prepare them for application to the body. The bands shall be heated uniformly by a process which will prevent oxidation (loose scale), and applied in a manner that will hold the formation of scale between the band and the band seat to a minimum.

3.5.4 Band seating. The diametrical clearance between the rotating band and band seat shall not exceed .006 inches, when the destructive method of testing is used or .009 inches when non-destructive method of testing is used.

3.6 Metal defects.

3.6.1 Body. The body, including the cavity, shall be free from cracks, splits, bursts, cold shuts, pipe, porosity, inclusions, folds, seams and other metal defects.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:51Z
Check the source to verify that this is the current version before use.

3.6.2 Body cavity. The body cavity shall be free of scale, fins, burrs, draw marks, laminations, imbedded foreign matter, pits and sharp edges.

3.6.3 Visual standards. Visual standards for metal defects of 3.6.2 will be established for each contract after the first month's production or whenever requested by the procuring activity (see 6.8).

3.7 Surface finish. The requirements for surface finish are as detailed on the applicable drawing.

3.8 Ultrasonic inspection. The projectile shall be free of any discontinuities equal or in excess of the inspection criteria within their respective zones when tested in accordance with 4.5.4. The inspection criteria are notches and holes in standard (see 9280420) which simulate larger actual cracks and piping flaws.

3.9 Proving ground functional test. The projectiles, when tested at excess pressure, shall show no loss or break up of metal components in flight.

3.10 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.11 Workmanship. The requirements for workmanship are as shown on the applicable drawing, in the applicable specifications and the following:

3.11.1 Surface preparation. Preparation of metal surfaces shall be as specified on the applicable drawings and in the applicable specifications referenced thereon. Where cleaning and preparation are required by component details, the finish specified shall apply to indicated surfaces regardless of contour configuration.

3.11.2 Painting. Painting shall comply with the requirements of the applicable drawing. All paint shall be dry to the touch before packing for shipment. Drying time prior to testing shall be in accordance with the applicable specification.

3.11.3 Threads. Threads shall be full and undamaged for the entire minimum length or depth as specified on the applicable drawing, and shall be in accordance with the provisions of Specification MIL-A-2550.

3.11.4 Base cover weld. The weld shall be sound, and the welding operation shall be so controlled that there will be adequate fusion between projectile body and base cover.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:51Z
Check the source to verify that this is the current version before use.

3.11.5 Burr. No part shall have a burr which might interfere with the assembly of the round or which might be injurious to personnel handling the item.

3.11.6 Foreign matter. No part or assembly shall contain dirt, grease, chips, rust, corrosion or other foreign matter.

3.11.7 Protection of rotating band. The machined rotating band shall be protected from being nicked, burred or otherwise damaged in subsequent handling prior to shipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspection. The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:51Z
Check the source to verify that this is the current version before use.

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawing	Quantity
Cover, Base	10535928	5
Blank, Rotating Band	7548993	5
Body, (Prior to Nosing)	10535927	5
Body, (Prior to Band Seat Machining and Threading Nose)	10535927	5
Body	10535927	5
Body Assembly	10535926	5
Projectile Metal Parts Assembly	10535925	45

4.3.2 Inspection to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:51Z
Check the source to verify that this is the current version before use.

TABLE I - First article inspection.

MIL-P-14824B (AR)

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Projectile Assemblies and Components	1 OF 2	See Below
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Cover, Base				
	Dwg. 10535928				
	Examination for Defects	5	100%	3.1	4.4.2.1
	Blank, Rotating Band				
	Dwg. 7548993				
	Mechanical Properties	5	100%	3.5.2	4.5.2.2.2
	Body (Prior to Nosing)				
	Dwg. 10535927				
	Examination for Defects	5	100%	3.6	4.4.2.2/4.5.1
	Body (Prior to Band Seat Machining and Threading Nose)				
	Dwg. 10535927				
	Examination for Defects	5	100%	3.1	4.4.2.3
	Ultrasonic Test	5	100%	3.8	4.5.4

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:51Z
Check the source to verify that this is the current version before use.

TABLE I - First article inspection.

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-14824B (AR)

PARAGRAPH	TITLE	SHEET OF	DRAWING NUMBER
	Projectile Assemblies and Components	2 2	See Below
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Body				
	Dwg. 10535927				
	Examination for Defects	5	100%	3.1	4.4.2.4
	Mechanical Properties				
	Tensile Strength	2	100%	3.4	4.5.2.2.1
	Body Assembly				
	Dwg. 10535926				
	Examination for Defects	5	100%	3.1	4.4.2.5
	Band Seating	5	100%	3.5.4	4.5.3
	Mechanical Properties - Rotating Band	5	100%	3.4.2	4.5.2.2.3
	Base Cover Weld Test	5	100%	3.11.4	4.5.5
	Projectile Metal Parts Assembly				
	Dwg. 10535925				
	Examination for Defects	5	100%	3.1	4.4.2.6
	Salt Spray Testing	5	100%	3.1	4.5.6.1
	Protective Coating Thickness	5	100%	3.1	4.5.6.2
	Functional Test (Proving Ground)				
	Dwg. 10535925				
	Ballistic Testing	40	100%	3.9	4.5.7

NOTES:

AMSMC rm 1570, 1 Feb 85 Repl ; DRSMC-QA (D) Form 160, 1 Aug 83, which may not used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:51Z
Check the source to verify that this is the current version before use.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, the following applies:

a. Heat treatment lots. Heat treatment lots shall contain heat treated bodies from not more than one heat of steel that have been heat treated in the same heat treatment equipment, at the same temperature, in one unchanged process. The heat treatment lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

b. Stress relieved lots. Stress relieved lots shall contain stress relieved bodies, from not more than one heat of steel, that have been cold shaped by the same process, utilizing the same tooling and equipment and stress relieved in the same furnace at the same temperature for the same length of time. The stress relieved lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

c. Miscellaneous lots. Small lots (such as bodies manufactured from abnormally small heats of steel or bodies that have become separated from previously delivered lots, etc.) may at the discretion of the supplier be grouped into miscellaneous heat treatment or stess relieved lots as the case may be. Such lots shall be identified by a miscellaneous lot code number and processed in accordance with either (a) or (b) above, as applicable. The miscellaneous lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

d. Projectile lots. When it has been determined that bodies of a heat treated or stress relieved lot have complied with the mechanical properties requirement of the applicable drawing, they may be grouped for further processing.

e. Shipping lots. Shipping lots shall consist of projectile assemblies produced by one supplier under one contract in one unchanged process, in accordance with the same drawings, revision, under conditions of a continuous production, free of interruptions other than those due to the end of shift day, or work week. Shipping lots shall be assigned a lot number in accordance with MIL-STD-1168. The shipping lots thus constituted shall be the basis for selection of samples for the proving ground functional test. Lot size shall be determined by the procuring contracting officer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:51Z
Check the source to verify that this is the current version before use.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:51Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-14824B (AR)

PARAGRAPH	TITLE		SHEET OF	DRAWING NUMBER
4.4.2.1	Base, Cover		1 1	10535928
				NEXT HIGHER ASSEMBLY 10535925
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Thickness		0.40%	3.1	Gage
Minor					
201	Diameter		0.65%	3.1	Gage
202	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:51Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET	OF	DRAWING NUMBER
4.4.2.2	Body (prior to nosing)			1 1	10535927
					NEXT HIGHER ASSEMBLY 10535926
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical 1	Metal defective cavity		100%	3.6	4.5.1 Visual Examination
Major	None defined				
Minor 201	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:51Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET OF	DRAWING NUMBER
4.4.2.3	Body (Prior to Band Seat Machining and Threading Nose)	1 1	10535927
			NEXT HIGHER ASSEMBLY: 10535926

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critcal					
1	Ultrasonic test		100%	3.8	4.5.4
2	Ultrasonic test stamp missing		100%	3.1	Visual
Major	None defined				
Minor	None defined				

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:51Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-14824B (AR)

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.4.2.4	Body		1 2		10535927
					NEXT HIGHER ASSEMBLY 10535926
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Perpendicularity of rear face with body		0.40%	3.1	Gage
102	Circular runout of pitch diameter of thread with body and front face		0.40%	3.1	Gage
103	Circular runout of ogive with body		0.40%	3.1	Gage
104	Circular runout of bourrelet with body		0.40%	3.1	Gage
105	Circular runout of boattail with body		0.40%	3.1	Gage
106	Diameter of rotating band seat		0.40%	3.1	Gage
107	Location of rotating band seat		0.40%	3.1	Gage
108	Width of rotating band seat		0.40%	3.1	Gage
109	Length from front of boattail to rear of rotating band seat		0.40%	3.1	Gage
110	Depth of scoring		0.40%	3.1	Visual
111	Scoring or knurling missing from rotating band seat		0.40%	3.1	Visual
112	Diameter of bourrelet, min.		0.40%	3.1	Gage
113	Diameter of body		0.40%	3.1	Gage
114	Pitch diameter of thread		0.40%	3.1	Gage
115	Minor diameter of thread max.		0.40%	3.1	Gage
116	Thickness through base		0.40%	3.1	Gage
117	Thickness of body wall min.		0.40%	3.1	Gage
118	Variation in wall thickness at ogive and body		0.40%	3.1	Gage

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:51Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 10535927
4.4.2.4	Body (continued)				NEXT HIGHER ASSEMBLY 10535926
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
119	Total length		0.40%	3.1	Gage
120	Length from front face to forward basic diameter		0.40%	3.1	Gage
121	Length from front face to rear basic diameter		0.40%	3.1	Gage
122	Length of effective thread, min.		0.40%	3.1	Gage
123	Length of thread stock, max.		0.40%	3.1	Gage
124	Mechanical properties Test Tension		Note (a)	3.4.1	4.5.2.2.1
Minor					
201	Diameter of thread chamfer		0.65%	3.1	Gage
202	Angle of thread chamfer		0.65%	3.1	Gage
203	Radii or chamfers missing or incorrect		0.65%	3.1	Visual
204	Diameter at rear of boattail		0.65%	3.1	Gage
205	Length from front face to rear of bourrelet		0.65%	3.1	Gage
206	Surface finish improper		0.65%	3.7	Visual
207	Foreign matter in cavity		0.65%	3.1	Visual
208	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES: (a) See paragraph 4.4.3.2.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:51Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 10535926
4.4.2.5	Body Assembly				NEXT HIGHER ASSEMBLY 10535925
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Metal defective		100%	3.6	Visual
2	Missing base plate, broken base plate weld or perforated base plate		100%	3.1	Visual
Major					
101	Circular runout of rotating band with body		0.40%	3.1	Gage
102	Diameter of rotating band		0.40%	3.1	Gage
103	Diameter of rotating band relief groove		0.40%	3.1	Gage
104	Location of rotating band		0.40%	3.1	Gage
105	Width of rotating band relief groove		0.40%	3.1	Gage
106	Gap between sidewall(s) at band seat and rotating band		0.40%	3.5.3	Visual
107	Weight		0.40%	3.1	Balance scale
108	Mechanical properties - rotating band		Note (a)	3.4.2	4.5.2.2.2
109	Rotating band seating		Note (b)	3.5.4	4.5.3

NOTES: Note (a) See paragraph 4.4.3.2.2.4. Note (b) See paragraph 4.4.3.3.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:51Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-14824B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Body Assembly (continued)	2 OF 2	10535926
			NEXT HIGHER ASSEMBLY 10535925

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Diameter of rotating band groove		0.65%	3.1	Gage
202	Length from rotating band groove to start of angle		0.65%	3.1	Gage
203	Length of rotating band flat		0.65%	3.1	Gage
204	Length total max		0.65%	3.1	Gage
205	Angle at front of rotating band		0.65%	3.1	Gage
206	Angle at rear of rotating band relief groove		0.65%	3.1	Gage
207	Width of rotating band		0.65%	3.1	Gage
208	Width of rotating band groove		0.65%	3.1	Gage
209	Location of base cover		0.65%	3.1	Gage
210	Depth of undercut front and rear of rotating band (if applicable)		0.65%	3.1	Visual
211	Width of undercut front and rear of rotating band (if applicable)		0.65%	3.1	Visual
212	Foreign matter in cavity		0.65%	3.1	Visual
213	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:51Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 10535925
4.4.2.6	Projectile, 155MM, H.E. M107 Metal Parts Assembly				NEXT HIGHER ASSEMBLY 9216352
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Pool of paint in cavity		100%	3.1	Visual
2	Metal defective cavity		100%	3.6	Visual
Major					
101	Diameter of bourrelet, max. (after painting)		100%	3.1	Gage
102	Rotating band damaged		0.40%	3.1	Visual
103	Foreign matter in cavity		0.40%	3.1	Visual
104	Protective coating thickness		Note (a)	3.1	4.5.6.2
105	Salt spray test		Note (a)	3.1	4.5.6.1
Minor					
201	Paint on threads		0.65%	3.1	Visual
202	Paint on rotating band bearing surface		0.65%	3.1	Visual
203	Protective coating in cavity inadequate or damaged		0.65%	3.1	Visual
204	Protective coating on exterior surfaces inadequate or damaged		0.65%	3.1	Visual
205	Anti-seize compound missing from thread		0.65%	3.1	Visual
206	Marking missing, incorrect or illegible		0.65%	3.1	Visual
207	Grommet missing		0.65%	3.1	Visual
208	Plug, shock attenuating missing		0.65%	3.1	Visual
209	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES: a. See paragraph 4.4.3.5.

AMSMC m 1570, 1 Feb 85 Repl DRSMC-QA (D) Form 160, 1 Aug 83, which may not used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:51Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.7	Projectile, 155MM, H.E., M107 Metal Parts Assembly Destructive Lot Acceptance	1	1	10535925
				NEXT HIGHER ASSEMBLY 9216352

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Government Proving Ground Ballistic Test (excess pressure)				
Critical 1	Components break-up or become disassembled inbore or in flight	Note (a)	100%	3.9	4.5.7.1
Major	None defined				
Minor	None defined				

NOTES:
(a) See paragraphs 4.4.4.2 and 4.4.4.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:51Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-14824B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9362569
4.4.2.8	Packing & Marking				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Pallet damaged (unserviceable)		0.65%	3.1	Visual
202	Improperly packed		0.65%	3.1	Visual
203	Strapping missing, loose or broken		0.65%	3.1	Visual
204	Marking missing, incorrect or illegible		0.65%	3.1	Visual/Manual
205	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:51Z
Check the source to verify that this is the current version before use.

4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection tables.

4.4.3.1 Steel quality. Records of test documentation shall be maintained and made available for Government review to provide evidence of compliance with the requirements of 3.2.

4.4.3.2 Mechanical properties.

4.4.3.2.1 Hardness.

4.4.3.2.1.1 Heat treatment lots. In order to determine uniformity of heat treatment, each body in the heat treatment lot (as defined by 4.4.1a) after heat treatment, but prior to final machining of the bourrelet diameter, shall be subjected to a hardness test with a Brinnel Hardness Tester or any other method approved by the procuring activity. Hardness impressions shall be made on the body bourrelet diameter and used as the basis for selecting the tension test samples of 4.4.3.2.2.1.

4.4.3.2.1.2 Stress relieved lots. Twenty-five (25) cold shaped bodies from the first batch of each stress relieve lot (as defined by 4.4.1b) processed through the stress relief furnace and 5 bodies from each one hours' production thereafter for the balance of the stress relieved lot shall be subjected to a hardness test to assess uniformity of the product.

4.4.3.2.1.3 Miscellaneous lots. Miscellaneous heat treatment lots shall be subjected to a hardness test in accordance with 4.4.3.2.1.1. Miscellaneous stress relieved lots shall be subjected to a hardness test in accordance with 4.4.3.2.1.2

4.4.3.2.2 Tension test.

4.4.3.2.2.1 Heat treatment lots. Two (2) projectile bodies one of the hardest and one of the softest as determined by the hardness test shall be subjected to this test. Failure of any tension test specimen to comply with the yield strength of elongation requirements of the applicable drawing shall be cause for rejection of the lot. At the option of the supplier a production control sample may be utilized to establish a hardness range. When a production control sample is used, the hardness range thus established shall be maintained throughout the heat treatment lot. The size of the sample shall be determined by the contractor. Projectile bodies which fall outside of the established range may be re-heat treated to fall within the range or new mechanical properties may be determined from newly selected projectile bodies which will represent the true extremes of hardness in the heat treatment lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:51Z
Check the source to verify that this is the current version before use.

4.4.3.2.2.2 Stress relieved lots. Two (2) projectile bodies, one of the hardest and one of the softest, as determined by the hardness test from the first twenty-five projectiles shall be subjected to this test. Failure of any tension test specimen to comply with the mechanical properties requirement of the applicable drawing shall be cause for rejection of the lot. When it has been determined that the mechanical properties comply and furnace temperature and time have been established, the furnace temperature and time shall be maintained throughout each stress relief lot. The furnace temperature may vary within twenty-five (25) degrees Fahrenheit but not below the minimum temperature of 700 degrees Fahrenheit and the minimum 30 minutes soak time. Furnace temperature and time will be recorded every hour to assess adequate control of stress relief equipment. The furnace temperature will be calibrated each eight (8) hours by an independent method.

4.4.3.2.2.3 Miscellaneous lots. Samples for tension test of miscellaneous heat treatment or stress relieved lots shall be selected in accordance with 4.4.3.2.2.1 or 4.4.3.2.2.2, as applicable. Failure of any tension test specimen to comply with the yield strength of elongation requirements of the applicable drawings shall be cause for rejection of the lot. Tension tests shall be performed on each miscellaneous heat of steel before being grouped into a miscellaneous lot.

4.4.3.2.2.4 Rotating band. One body assembly shall be selected from each 8-hour production or fraction thereof from each banding machine in operation and subjected to this test to determine compliance with 3.4.2. Failure to comply shall be cause for rejection of the assemblies represented by the sample.

4.4.3.3 Band seating.

4.4.3.3.1 Tightness.

4.4.3.3.1.1 Nondestructive method. Five body assemblies from each 1-hour production or fraction thereof per banding machine in operation shall be subjected to this test. Failure of one or more bands to comply with 3.5.4 shall be cause for rejection of the assemblies represented by the sample.

4.4.3.3.1.2 Destructive method. When the nondestructive equipment is not available, one body assembly from each 8-hour production or fraction thereof from each banding machine in operation shall be subjected to this test to determine compliance with 3.5.4. Failure to comply shall be cause for rejection of the assemblies represented by the failing sample.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:51Z
Check the source to verify that this is the current version before use.

4.4.3.3.2 Sidewall contact. Rotating bands removed as specified under 4.4.3.2.2.4 shall be examined to determine compliance with 3.5.3. Failure to comply shall be cause for rejection of assemblies represented by the sample.

4.4.3.4 Base cover weld. Five body assemblies from each 1 hours production or fraction thereof per welding machine in operation shall be subjected to this test. Failure of the weld to comply with the requirements of 3.11.4 shall be cause for rejection of the production represented by the sample and shall be screened 100 percent.

4.4.3.5 Protective coating - salt spray. The sample size and acceptance criteria shall be as specified in the applicable specification. If failure occurs, all items processed since last acceptance shall be rejected and corrective action taken.

4.4.3.5.1 Salt spray. After qualification in accordance with TT-C-490, two (2) salt spray samples shall be taken each week, one during the first 30 minutes of production and a second at any other time. Failure to comply with the requirements of the applicable drawing shall be cause for taking the corrective action prescribed in TT-C-490.

4.4.4 Proving ground functional tests. The following proving ground tests will be performed by the Government.

4.4.4.1 First article sample. Forty (40) projectiles shall be randomly selected from the first article sample for the excess pressure test. Failure of any projectile assembly to comply with 3.9 shall be cause for rejection of the first article sample.

4.4.4.2 Initial production lot samples. After the first article sample has met the ballistic criteria, 30 projectiles shall be randomly selected from each of the first three production lots for the excess pressure test. Failure of any projectile assembly to comply with 3.9 shall be cause for rejection of the initial production lot.

4.4.4.3 Production lot samples. After three (3) consecutive initial production lots have met the ballistic criteria, 10 projectiles shall be randomly selected from each production lot for the excess pressure test. Failure of any projectile assembly to comply with 3.9 shall be cause for rejection lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:51Z
Check the source to verify that this is the current version before use.

4.4.5 Tool control dimensions.

a. Dimensions marked tool control shall be gaged at the beginning or end of each working day (24 hrs), during First Article Inspection and for the first component after each tooling change, adjustment or repair. Where destruction of the component is necessary to gage such dimensions, measurements of the tool may be substituted providing the contractor has established correlation between the tool dimensions and the component dimensions prior to the start of production.

b. When the contractor has satisfied (a) above, visual inspection of the tool control dimension(s) will be permitted. Visual inspection shall be performed every two (2) hours for each machine in operation. If any tool control dimension(s) is in doubt during visual inspection, the dimension(s) shall be immediately gaged to determine acceptability. Corrective action will immediately be instituted by the contractor whenever a defective tool control dimension is found.

4.4.6 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the 'Paragraph Reference/Inspection Method' column in the table starting in paragrapsh 4.4.2.1. The contractor shall submit for approval of inspection equipment designs in accordance with the terms of the contract. See section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection.

4.5.1 Visual examination. The internal cavity of the projectile shall be inspected visually prior to nosing using peripheral lighting to assure conformance with the applicable requirements. Visual standards in accordance with 3.6.3 shall be set up for use during this inspection. Any warhead failing to meet the applicable requirements of 3.6 shall be classed defective and removed from the lot.

4.5.2 Mechanical properties.

4.5.2.1 Hardness. The hardness test shall be conducted in accordance with ASTM Method E 103 and the following:

a. The surface of the body in the area to be hardness tested shall be prepared by machining or grinding to a sufficient depth, .015 inches minimum, to remove scale and get below the decarburization zone.

b. Hardness tester periodic verification during each operating shift shall be performed using Brinell test standards manufactured in accordance with ASTM Method E-10 in lieu of production samples.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:51Z
Check the source to verify that this is the current version before use.

c. All hardness test measurements shall be automatically displayed.

d. The hardness tester shall use a standard 10 MM Brinell ball. The body shall be securely supported so that no rocking or shifting of the body will occur during the test. The tester shall apply a preload followed by application of the full load. The preload shall not exceed 10% of the major load of 300Kg. The amount of penetration of the indentor (10MM Ball) during that time period when the major load is applied including the dwell time at the end of the load application shall be automatically measured. The tester shall be provided with a means to compensate for any flexing of the test item that may occur during application of the major load. The hardness number will be displayed at the conclusion of the test.

4.5.2.2 Tension test.

4.5.1.2.1 Body. From the samples provided by the heat treat or stress relief lots two tension specimens conforming to the largest obtainable standard "R" type specimen, as prescribed by Federal Test Method Standard No. 151, shall be taken longitudinally from the opposite side of the body at midwall, in such a manner that the center of the gage length of the machined test bar will fall in line coincident with the center of the rotating band seat. The number of specimens and their locations are for the convenience of the contractor and Government and shall not restrict the contractor or Government from testing additional specimens at other locations to determine that the mechanical properties of the projectile body meet the drawing requirments. The criteria for performing this test shall be prescribed by ASTM Method E-8; except that elongation shall be measured to the nearest thousandth (0.000) prior to conversion into percent elongation. Present elongation shall be recorded to the nearest tenth (0.0 percent).

4.5.2.2.2 Mechanical properties of rotating band blanks. The band blank shall be cut from front to rear and flattened, without hammering which may further cold work the material, and the largest obtainable flat type specimen shall be cut from the band, the length of which shall provide a one inch gage length, plus adequate material at each end to accommodate the grips of the tensile testing machine. This method should yield a specimen with a 0.500 inch gage width. In any event, the gage cross section shall be determined to the nearest 0.001 square inch and utilized to calculate the tensile and yield strengths, and elongation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:51Z
Check the source to verify that this is the current version before use.

4.5.2.2.3 Mechanical properties of disassembled rotating band. The sample provided by 4.4.3.2.2.4 shall have the rotating band removed and carefully flattened without hammering so as to hold a minimum any additional cold working. The specimen blank shall be machined so that all serration marks from the band seat are removed. A test specimen having one inch gage length and one-half inch gage width, and grip ends of sufficient length to accommodate the tensile machine shall be removed from the blank and tested in accordance with the procedure prescribed by ASTM Method E-8.

4.5.3 Band seating.

4.5.3.1 Tightness.

4.5.3.1.1 Non-destructive method. The samples provided by 4.4.3.3.1.1 shall be tested utilizing the equipment and procedures prescribed in Frankford Arsenal Detroit Testing Machine Manual (see 6.10).

4.5.3.1.2 Destructive method. The body assembly selected in accordance with 4.4.3.3.1.2 should not be processed through the band machining line, although each band should have a light surface cut to true up the surface prior to measuring. The diameter of each band shall be measured on 3 locations, 60 degrees apart, on the points along the cylindrical portion of their length, near the front and rear edges. This comprises 6 diametrical measurements per band. These points of measurement shall be marked on the bands and on the projectile body adjacent thereto in such a manner as to permit proper body and band orientation after removal. The bands shall be carefully removed so as to keep deformation to a minimum. After removal, the thickness of the band shall be measured at each of the marked points (a total of 12 measurements) and the diameter of the band seat corresponding to these points shall also be measured. Add to each of the diameters of the band seat the sum of the corresponding 2 thicknesses of the band; the results obtained shall then be subtracted from the previously determined diameters over the band.

4.5.3.2 Sidewall contact. The samples provided in accordance with 4.4.3.2.2.4 shall be sectioned by cutting through two planes normal to the longitudinal axis one-half inch forward and rear of the rotating band. The cylindrical portion containing the rotating band shall be sectioned approximately in half through a plane parallel to its longitudinal axis. One of the two halves shall be used for establishment of mechanical properties of the band; the other section shall be sectioned into two parts, thus exposing four surfaces each of which reveals a view of the band swaged into the band seat. The exposed surfaces shall be subjected to a light grinding operation to remove metal smear effects of saw cuts prior to measurements for depth of sidewall contact to determine compliance with the requirements of 3.5.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:51Z
Check the source to verify that this is the current version before use.

4.5.4 Ultrasonic inspection. The projectile body shall be ultrasonically inspected prior to band seat machining and threading nose using equipment and procedures in accordance with 4.5.4.1. The projectile shall be inspected with ultrasonics throughout all of the material. Ultrasonic energy shall be used at right angles to assure that orientation does not prevent detection of a defect. One standard in accordance with 4.5.4.3 shall be used to check the effectiveness of the inspection process. Any projectile failing to meet the requirements of 3.8 shall be classed defective and removed from the lot.

4.5.4.1 Equipment and procedures. The ultrasonic inspection shall be performed in accordance with Dwg. 9280419 with the use of a liquid to assure a reliable coupling to transmit ultransonic energy. Only immersion testing shall be used.

Using ultrasonic energy at right angles requires at least two separate sources of ultrasonic energy. This means that in the nose, ogive and bourrelet regions, ultrasonic energy shall move within the walls of the projectile with separate significant axial and circumferential components of velocity to detect flaws located anywhere on the outside, inside or within the walls. For the base region, significant ultrasonic velocity components shall separately be radial and circumferential.

Criteria for detection of cracks are notches which simulate actual larger cracks. Details are contained on Drawing 9280420. Criteria for detection of piping in the base region consists of two drilled holes oriented parallel to the major axis of the projectile.

All notches and holes machined on the standards are targets for rejection. That is, any projectile that sends back an echo signal that is greater than or equal to the signal from the standard shall be rejected.

4.5.4.2 Verification and operation procedures and equipment. Prior to initiation of ultrasonic inspection, the contractor shall submit for written approval one copy of his proposed procedures, including descriptions of equipment (contractor design) and operation procedures (see 6.3). No change to the procedure shall be made after the original approval without submission and approval of the proposed changes. As a minimum the adequacy of the inspection process shall be checked by using the standards at the start and finish of each production shift or change in operator, as well as at four (4) hour intervals during production. All channels shall properly indicate a rejection signal when encountering the notches and holes mentioned in 3.8. If the inspection process is determined to be inadequate, the condition shall be corrected and all items inspected since the

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:51Z
Check the source to verify that this is the current version before use.

last acceptable inspection check shall be reinspected. The verification check may be conducted by the equipment operator. Any adjustment required shall be performed by a Level II certified individual (See 4.5.4.4).

Equipment used shall provide for a positive activation of a buzzer or alarm as well as a light system for each channel to alert the operator of the signal indications that are the criteria for rejection. A log book shall be maintained by the operator of each system which as a minimum, documents the use of standards, number of items inspected, number of rejects by zone, changing of transducers, instrumentation, repairs, reasons for equipment breakage, if any, and changing of qualified operators or inspectors. This log book shall be made available when requested for Government review.

4.5.4.3 Ultrasonic standards. Ultrasonic inspection standards shall be provided by the contractor for use during the ultrasonic inspection. These standards shall be machined in accordance with drawing 9280420 and shall be inspected for all drawing requirements to determine their acceptability for use. Government verification of all inspections performed on the standards is required. In addition, a certificate of conformance for the standards shall be submitted to the Government, prior to use of the standards.

4.5.4.4 Operator qualifications. For operation of ultrasonic equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level I. For set-up and calibration of ultrasonics equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level II.

4.5.5 Base cover weld. The test of the base cover weld shall be performed as follows: visually inspect the base cover and select the area that appears to have the weakest weld; insert a 5/8" chisel in this area and apply force by use of a one pound hammer and an approximate one foot stroke. If the weld breaks loose after 2 strokes, the body assembly will be classed defective. If the weld is found to be secure, the burr made by the chisel may be knocked back into position and accepted.

4.5.6 Protective coating. The procedures for testing the cleaning of components and the application of protective coating shall be in accordance with the requirements of the applicable drawing and the specification listed thereon. (See 6.11).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:51Z
Check the source to verify that this is the current version before use.

4.5.6.1 Salt spray testing. Salt spray testing shall be accomplished on production projectiles or sectioned projectiles that were acceptable at the time of painting and were sectioned subsequent to painting. If sectioned projectiles are used, sectioning shall be accomplished to allow testing of all major areas of the projectiles on an alternating basis. The salt spray test shall be performed utilizing the equipment and procedures specified by ASTM Method B-117 and TT-C-490 with the exception that test samples shall be selected in accordance with 4.4.3.5.1 and test samples shall be dried in accordance with Drawing 10535925.

4.5.6.2 Protective coating - thickness. Protective coating thickness shall be accomplished using commercially available equipment.

4.5.7 Proving ground functional tests.

4.5.7.1 Excess pressure. The ballistic test projectiles inert loaded and dummy fuzed will be fired in an acceptable M126 series cannon based upon current gun tube condemnation criteria. The propellant charge XM119E4 shall be adjusted to meet 54,000 psi plus or minus 2,000 psi. Photographic equipment (smear camera) will be set up near the gun muzzle to record the flight of the projectile. Visual observation will be made to assure projectile impacts in target area. Velocity and chamber pressures will be recorded.

5. PACKAGING

5.1 Packing - Level C. Minimum military pack.

5.1.1 Packing. Projectile metal parts assemblies including lifting plug and grommet shall be palletized in accordance with drawing 9362569.

5.2 Marking. Marking shall be in accordance with dwg. 9362569

6. NOTES

6.1 Intended use. The item covered by this specification is intended for use on the Projectile 155MM, HE, M107.

6.2 Acquisition requirements. See MIL-A-48078.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:51Z
Check the source to verify that this is the current version before use.

6.3 Submission of inspection procedures and equipment designs for approval. See MIL-A-48078. Submit equipment designs, as required, to: Commander, US Army Armament, Munitions and Chemical Command, ATTN: AMSMC-QAR-I (D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.4 Drawings. Drawings listed in section 2 of this specification under the heading US Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal, US Army Research and Development Command (ARRADCOM) or US Army Armament Research and Development Center (ARDC) drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

6.6 Submission of test data and ammunition data cards. In addition to the normal distribution of records, when the projectile is procured by the US Army Armament Munitions and Chemical Command, one copy of all ballistic test data and ammunition data cards shall be forwarded to each of the following: Commander, ARDEC, ATTN: SMCAR-CCH-P and AMSMC-QAR-Q (D), Picatinny Arsenal, NJ 07806-5000, and Commander, ARRCOM, ATTN: AMSMC-QAD, Rock Island, IL 61299-6000.

6.7 Visual examination qualification. When compliance with the applicable requirements is in doubt as a result of visual examination, the characteristics may be measured or gaged to determine acceptability.

6.8 Visual standards. Visual standards to be established at contractor's plant in compliance with 3.6.3.

6.9 Alternative testing procedures. Alternative testing procedures may be used by the contractor when such testing and procedures provide as a minimum the quality assurance provisions specified herein. Prior to applying alternative procedures, the contractor shall describe them in a written proposal and shall demonstrate for the approval of the responsible technical agency that their effectiveness is equal to, or better than, the quality assurance provisions specified herein.

6.10 Band seating. The operating procedures for use of Detroit Testing Machine as prepared by the Inspection Equipment Branch, US Army Armament Research, Development and Engineering Center, AMSMC-QAR-I (D), Picatinny Arsenal, NJ 07806-5000.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:51Z
Check the source to verify that this is the current version before use.

6.11 Protective coating thickness. In checking the thickness of protective coating the "Dermitron" by Unit Process Assemblies Inc. 53-15 37th Ave., Woodside, NY 11377, has been proven to be satisfactory.

6.12 Steel quality.

a. It is the responsibility of the manufacturer of the projectile metal parts to select a fully-killed carbon steel of a grade compatible with the abilities of the equipment and the methods of processing the metal within the operations of the shops of the forger and machiner to produce the projectile to the dimensions and mechanical properties and metal integrity specified as an end item requirement.

b. The supplier of the selected steel raw material is responsible only for furnishing a metal composition, grade, and type with dimensions, weight, soundness and freedom from metallurgical or mechanical defect limitations identifiable with the standard ANSI grade selected by the projectile manufacturer and complying with the specific ASTM designation and tests selected by the projectile manufacturer.

c. In the application of carbon steel, blooms, billets, slabs and bars, forging quality processing techniques adapted to use of the selected steel should be applied with the expectation that there will be variables in the metal cross-sectional dimensions in weight per unit length, in the hardenability variance from heat to heat, in the mechanical condition of the surface, and in the straightness of the raw material. Economic advantages will be realized if the process and equipment within the shops of the forger and the machiner have been engineered in terms of equipment sequence of operations, tooling designs, handling devices, and processing techniques having capabilities of compensating for such variables.

6.13 Heat treatment. Heat treatment is understood to be any thermal process which brings about a modification in the mechanical and metallurgical properties of the metal involved.

6.14 Subject term (key word) listing.

Ammunition
Ballistic testing (inert loaded) metal parts security
Military Specifications
Projectile; 155MM, M107
Rotating Band Seating (band seating testing machine)

Custodian
Army - AR

Preparing activity:
Army - AR

(Project No. 1320-AB09)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:51Z
Check the source to verify that this is the current version before use.